

# ENTITY ANALYSIS REPORT

## Core System Architecture & Data Modeling

July 2026

### 1. Executive Summary

This document presents a comprehensive analysis of the core data entities required for the foundational system architecture. The objective of this report is to outline key data structures, their attributes, and interconnecting relationships to ensure optimal data integrity, scalability, and system performance.

**Strategic Objective:** To define a standardized, robust entity-relationship model that supports cross-functional business operations, seamless data migrations, and advanced analytics.

## 2. Core Entity Identification

The system's data architecture is built upon several foundational entities that capture the primary business domains. The initial phase of this project identifies the following major entities:

- **User Profile:** Represents all individuals interacting with the platform (Customers, Admins, Support).
- **Inventory Item:** Represents the products or services managed within the system.
- **Transaction Record:** Represents the historical log of interactions between Users and Inventory Items.

## 3. Detailed Attribute Breakdown

### 3.1 User Profile Entity

The User Profile entity centralizes identity and access management data.

| ATTRIBUTE NAME | DATA TYPE    | DESCRIPTION / CONSTRAINTS                                      |
|----------------|--------------|----------------------------------------------------------------|
| UserID         | UUID         | Primary Key. Auto-generated unique identifier.                 |
| EmailAddress   | VARCHAR(255) | Unique constraint. Used for authentication and communications. |
| RoleID         | INT          | Foreign Key linking to access control matrix.                  |
| AccountStatus  | ENUM         | Values: Active, Suspended, Pending Verification.               |
| CreatedAt      | TIMESTAMP    | Default: CURRENT_TIMESTAMP.                                    |

## 3.2 Inventory Item Entity

This entity tracks all tangible or digital assets available within the operational ecosystem.

| ATTRIBUTE NAME | DATA TYPE     | DESCRIPTION / CONSTRAINTS                         |
|----------------|---------------|---------------------------------------------------|
| ItemID         | UUID          | Primary Key.                                      |
| SKU_Code       | VARCHAR(50)   | Unique constraint. Stock Keeping Unit identifier. |
| BasePrice      | DECIMAL(10,2) | Current financial value of the item.              |
| StockQuantity  | INT           | Available units in inventory. Must be $\geq 0$ .  |

## 4. Entity Relationships & Mapping

To maintain strict referential integrity, the system utilizes standard relational mapping conventions. The primary pathways include:

### User Profile to Transaction Record (1:N)

A single User Profile can be associated with multiple Transaction Records over its lifecycle. The `UserID` serves as a Foreign Key within the Transaction entity, ensuring all actions are traceable to a specific identity.

### Transaction Record to Inventory Item (M:N)

Transactions often involve multiple items, and a single item can be part of multiple transactions. This many-to-many relationship is resolved via an associative entity (e.g., `TransactionLineItem`) bridging the gap and storing quantity and price-at-time-of-sale data.

## 5. Recommendations & Next Steps

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With the core entities defined, the next phases of the project will focus on the technical implementation. Recommended actions include:

- **Database Normalization:** Review all tables to ensure 3NF (Third Normal Form) compliance to eliminate data redundancy.
- **DDL Generation:** Draft the initial SQL Data Definition Language scripts to provision the development database environments.
- **Index Optimization:** Identify frequently queried columns (e.g., EmailAddress, SKU\_Code) and design an appropriate indexing strategy to minimize latency.